

Adaptive Motion–Appearance Fusion for Robust Multi-Object Tracking with Liquid Time-Constant Motion Prediction

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Project repository: <https://github.com/khangkaka066/ByteTrack-DMA-LTC-Motion-Tracker>

Abstract—Multi-object tracking (MOT) remains challenging in crowded scenes where nonlinear motion, occlusion, and appearance ambiguity frequently cause identity switches and fragmented trajectories. Although ByteTrack provides a strong and efficient tracking-by-detection baseline, its association reliability is still constrained by linear Kalman motion prediction and static motion–appearance fusion. This paper presents an adaptive extension of ByteTrack that improves motion robustness and identity association through two complementary components. First, a Liquid Time-Constant/Closed-form Continuous-time (LTC/CfC) motion predictor is introduced as a residual correction module over Kalman prediction, enabling the tracker to model nonlinear temporal dynamics while preserving the stability of the original filter. Second, a Dynamic Motion–Appearance (DMA) fusion module learns pair-wise association weights between motion cost and Re-identification cost for each track–detection candidate. Instead of using fixed fusion weights, the proposed DynamicWeightNet estimates context-aware reliability from motion uncertainty, appearance consistency, detection confidence, and track history. Experiments on SportMOT and MOT17 show substantial improvements over representative baselines, achieving 78.35 HOTA and 98.12 MOTA on SportMOT, and 78.19 HOTA, 90.32 MOTA, and 87.63 IDF1 on MOT17. These results suggest that residual motion prediction and adaptive motion–appearance fusion are effective for improving identity consistency in complex tracking scenarios.

Index Terms—Multi-object tracking, ByteTrack, Kalman filter, Liquid Time-Constant network, Re-identification, data association, adaptive fusion.

I. INTRODUCTION

Multi-object tracking (MOT) aims to localize multiple objects in video sequences while maintaining consistent identities over time. It is a fundamental task in computer vision with applications in intelligent surveillance, autonomous driving, sports analytics, robotics, and human behavior understanding. Despite significant progress in modern tracking-by-detection systems, robust MOT remains difficult in crowded scenes due to long-term occlusion, nonlinear motion, similar appearances among targets, and unreliable detections.

Tracking-by-detection has become the dominant paradigm for online MOT. In this framework, an object detector produces frame-level bounding boxes, and a tracker links detections across time. ByteTrack [1] is a representative and highly

effective method in this category. Its key insight is to associate almost every detection box, including low-confidence detections, thereby reducing missed associations and improving trajectory continuity.

However, ByteTrack still depends heavily on Kalman filter prediction for motion modeling. While the Kalman filter is efficient and stable, it is inherently limited by a linear motion assumption. This assumption becomes less reliable when targets accelerate, abruptly change direction, interact with other objects, or remain occluded for several frames. Moreover, in trackers that incorporate appearance cues, motion and Re-identification (ReID) costs are often fused using fixed weights. Such a static strategy cannot reflect the fact that motion and appearance reliability varies across scenes, frames, and individual track–detection candidates.

To address these limitations, this paper proposes an adaptive extension of ByteTrack with two main components. First, we introduce an LTC/CfC residual motion predictor that refines Kalman prediction without replacing the original motion model. Second, we propose Dynamic Motion–Appearance (DMA) fusion, a lightweight neural module that predicts pair-wise weights between motion and appearance costs during association.

The main contributions of this paper are summarized as follows:

- We propose an LTC/CfC residual motion predictor that enhances Kalman-based tracking under nonlinear motion while preserving the stability and efficiency of ByteTrack.
- We introduce DMA fusion, which learns adaptive association weights for each track–detection pair instead of relying on fixed motion–appearance weighting.
- We design a compact reliability feature representation that captures motion uncertainty, appearance consistency, detection quality, and track history.
- We evaluate the proposed method on SportMOT and MOT17, demonstrating strong improvements in HOTA, MOTA, and IDF1 over representative baselines.

II. RELATED WORK

A. Tracking-by-Detection

Tracking-by-detection has become the dominant paradigm for modern multi-object tracking (MOT). In this framework, object detectors first localize targets in each frame, followed by a data association stage that links detections across time. Among representative approaches, DeepSORT incorporates appearance embeddings to improve identity preservation, while FairMOT jointly learns object detection and Re-Identification (ReID) features within a unified network. ByteTrack further advances this paradigm by introducing BYTE association, which leverages both high-confidence and low-confidence detections instead of discarding unreliable observations. This simple yet effective strategy significantly reduces false negatives and achieves state-of-the-art performance on multiple MOT benchmarks.

Despite its success, ByteTrack still relies on a linear Kalman Filter for motion prediction and uses hand-crafted association rules. Consequently, its performance may degrade under non-linear object motion, long-term occlusion, or crowded scenes where appearance ambiguity frequently occurs.

B. Motion Modeling for Multi-Object Tracking

Motion estimation plays a fundamental role in maintaining trajectory consistency. Traditional MOT methods generally employ Kalman Filters due to their computational efficiency and robustness under approximately linear motion. However, the constant-velocity assumption often becomes invalid in real-world scenarios involving abrupt acceleration, camera motion, or complex interactions among multiple objects.

Recent studies have explored deep learning-based motion models to overcome these limitations. Recurrent neural networks (RNNs), LSTMs, GRUs, and more recently Continuous-time Neural Networks such as Liquid Time-Constant (LTC) and Closed-form Continuous-time (CfC) networks have demonstrated stronger capability in modeling nonlinear temporal dynamics. These models capture long-term dependencies and continuously evolving trajectories that cannot be represented adequately by linear dynamical systems.

Instead of completely replacing the Kalman Filter, several recent approaches predict a residual motion correction on top of the Kalman estimate, preserving the stability of classical filtering while improving trajectory prediction under complex motion patterns. This residual-learning strategy motivates the motion prediction module proposed in our work.

C. Motion and Appearance Fusion

Reliable data association usually combines geometric motion cues with appearance similarity extracted by ReID models. Existing trackers often fuse these two sources using manually designed thresholds or fixed weighting coefficients. Examples include DeepSORT and BoT-SORT, where motion distance and appearance similarity are combined through empirically selected hyperparameters.

However, the relative importance of motion and appearance varies considerably across tracking scenarios. During long-term occlusion, appearance features may become more informative, whereas in crowded scenes with visually similar targets, motion continuity often provides stronger association cues. Fixed fusion strategies therefore cannot adapt to changing tracking conditions.

Several recent methods attempt to learn adaptive association mechanisms using neural networks or attention-based architectures. These approaches demonstrate that dynamically estimating the confidence of different cues can improve association robustness and reduce identity switches. Inspired by this observation, our method learns frame-wise adaptive fusion weights from multiple contextual features, enabling more flexible integration of motion and appearance information.

III. PROPOSED METHOD

A. Problem Formulation

Given a video sequence $\mathcal{V} = \{I_t\}_{t=1}^T$, multi-object tracking aims to estimate a set of object trajectories $\mathcal{T} = \{\tau_k\}_{k=1}^K$, where each trajectory τ_k contains temporally ordered bounding boxes associated with a consistent identity. At each frame t , the detector produces a set of detections $\mathcal{D}_t = \{d_j^t\}_{j=1}^{N_t}$. The tracker associates these detections with existing tracks by minimizing an association cost matrix.

For a track i and detection j , the proposed method computes a fused association cost:

$$C_{ij} = w_{ij}^m C_{ij}^m + w_{ij}^r C_{ij}^r, \quad (1)$$

where C_{ij}^m is the motion cost, C_{ij}^r is the appearance cost, and w_{ij}^m, w_{ij}^r are adaptive reliability weights predicted for each track-detection pair. The weights satisfy:

$$w_{ij}^m + w_{ij}^r = 1. \quad (2)$$

B. Overview

The proposed tracker extends ByteTrack with two complementary modules: an LTC/CfC motion residual predictor and a Dynamic Motion-Appearance fusion module. The overall pipeline is:

$$\text{YOLOX} \rightarrow \text{Kalman+LTC} \rightarrow \text{ReID} \rightarrow \text{DMA} \rightarrow \text{BYTE Association}. \quad (3)$$

The main design principle is to preserve the efficiency and simplicity of ByteTrack while improving the components most responsible for identity stability: motion prediction and data association.

C. Kalman-Based Baseline

ByteTrack represents each target state as:

$$\mathbf{x} = [x, y, a, h, \dot{x}, \dot{y}, \dot{a}, \dot{h}]^T, \quad (4)$$

where (x, y) denotes the bounding box center, a is the aspect ratio, h is the bounding box height, and the remaining variables represent corresponding velocities.

The Kalman prediction step is given by:

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}, \quad (5)$$

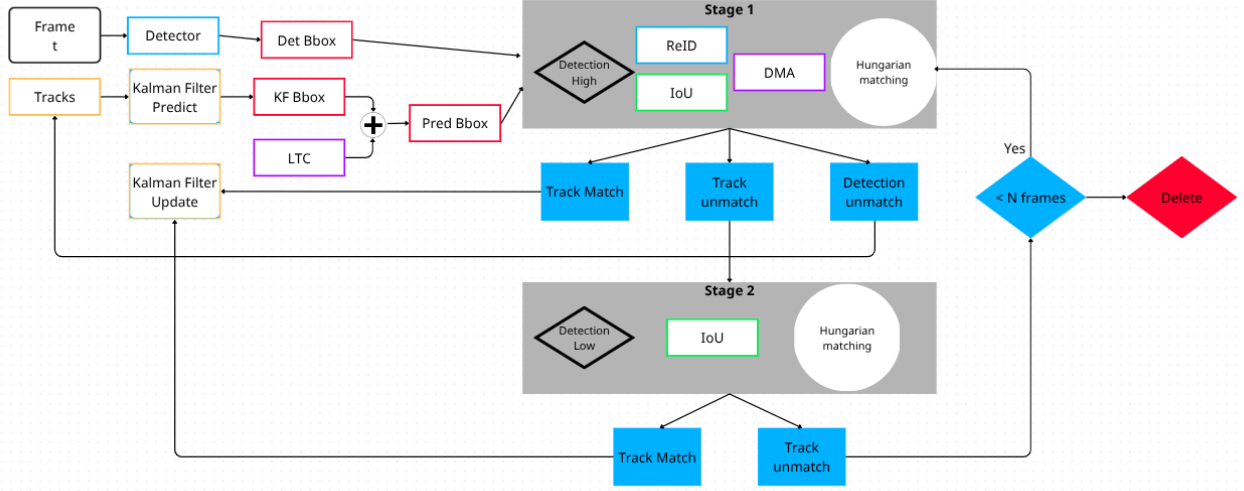


Fig. 1: Overall architecture of the proposed ByteTrack + LTC + DMA framework. The detector produces frame-level bounding boxes, the Kalman filter provides baseline motion prediction, the LTC/CfC module refines motion with residual correction, and DMA adaptively fuses motion and appearance costs before BYTE association.

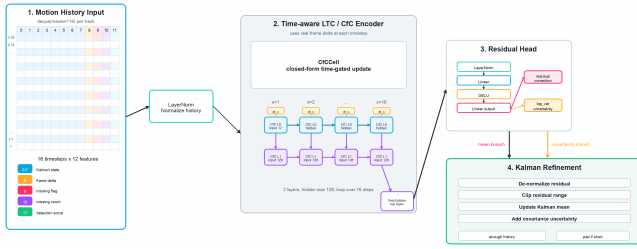


Fig. 2: Residual motion prediction with the LTC/CfC module. Instead of replacing Kalman prediction, the learned module estimates a residual correction from track history and refines the predicted measurement state.

$$\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^\top + \mathbf{Q}_t. \quad (6)$$

Although this formulation is efficient, it assumes approximately linear motion between adjacent frames. This assumption may fail in crowded scenes or sports videos where objects frequently accelerate, turn, or interact.

D. LTC/CfC Residual Motion Prediction

To improve motion prediction, we introduce an LTC/CfC residual predictor. Let $\hat{\mathbf{z}}_t^{KF}$ denote the Kalman-predicted measurement state. The LTC module predicts a residual correction:

$$\Delta \mathbf{z}_t^{LTC} = f_\theta(\mathcal{H}_t), \quad (7)$$

where $\mathcal{H}_t$ is the motion history of a track and f_θ is the LTC/CfC predictor. The refined prediction is:

$$\hat{\mathbf{z}}_t = \hat{\mathbf{z}}_t^{KF} + \Delta \mathbf{z}_t^{LTC}. \quad (8)$$

This residual formulation has two advantages. First, it preserves the stability and interpretability of Kalman prediction.

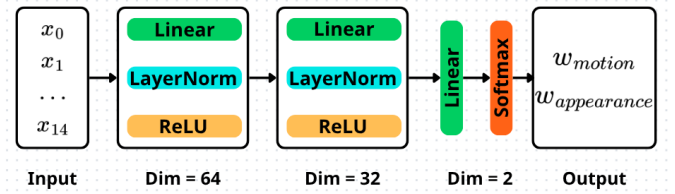


Fig. 3: Dynamic Motion-Appearance fusion. For each track-detection pair, DMA extracts reliability features and predicts adaptive weights for motion and ReID costs.

Second, it allows the model to learn nonlinear temporal corrections from data. The predictor also estimates uncertainty through a log-variance output, which can be used to adjust the predicted covariance.

E. Dynamic Motion-Appearance Fusion

Association quality depends on the reliability of both motion and appearance cues. In crowded scenes, either cue may become unreliable. DMA addresses this by predicting pairwise fusion weights. For each track-detection pair (i, j) , DMA extracts a feature vector:

$$\mathbf{f}_{ij} = [\mathbf{f}_{ij}^{motion}, \mathbf{f}_{ij}^{appearance}, \mathbf{f}_j^{detection}, \mathbf{f}_i^{history}]. \quad (9)$$

The DynamicWeightNet then predicts:

$$[w_{ij}^m, w_{ij}^r] = \text{softmax}(g_\phi(\mathbf{f}_{ij})). \quad (10)$$

The final fused association cost is computed using Eq. (1). This formulation allows the tracker to assign higher motion weight when appearance is ambiguous, and higher appearance weight when motion prediction is uncertain.

F. Feature Representation

The DMA input consists of 15 features grouped into four categories.

a) *Motion features.*: Motion features include IoU between the predicted bounding box and detection, motion cost, normalized Mahalanobis distance, covariance trace, covariance mean, velocity magnitude, and time since last update.

- IoU: The ratio of the intersection area to the total area of the two bounding boxes minus the intersection area. A higher IoU indicates that the two bounding boxes are closer to each other.

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (11)$$

- Motion Cost: Motion cost: an estimate of the difference between a track's predicted position and the current detection. It is inversely correlated with IoU (11); the lower the value, the better the two bounding boxes match.

$$motion_{cost} = 1 - IoU \quad (12)$$

- Mahalanobis distance: In addition to using IoU, we also employ the Mahalanobis distance [?] to characterize the DMA. Unlike IoU, which relies solely on geometry, Mahalanobis normalizes the deviation between predicted points and detections based on the track's uncertainty covariance. This enables DMA to distinguish between a "small deviation with an unstable track" and a "small deviation with a stable track."

For track i with a Kalman state projected into the measurement space $\hat{\mathbf{z}}_i \in \mathbb{R}^4$ (corresponding to x, y, a, h) and innovation covariance S_i , the squared Mahalanobis distance with respect to detection j (measurement $\mathbf{d}_j$) is calculated.:

$$\mathbf{D}_{ij}^2 = (\mathbf{d}_j - \hat{\mathbf{z}}_i)^\top S_i^{-1} (\mathbf{d}_j - \hat{\mathbf{z}}_i) \quad (13)$$

Since $\mathbf{D}_{ij}^2$ does not have a fixed upper bound, we normalize it using the 95% chi-square threshold with 4 degrees of freedom (corresponding to the dimensionality of $\hat{\mathbf{z}}_i$):

$$distance_{manaholobis} = \text{clip} \left(\frac{\mathbf{D}_{ij}^2}{\chi_{0.95,4}^2}; 0, 1 \right) \quad (14)$$

- Covariance trace & Covariance mean: While the Mahalanobis distance (14) measures the fit between a prediction and a specific detection, these two features measure the track's intrinsic uncertainty, independent of the detection under consideration. The trace of the covariance matrix represents the total variance across the state dimensions (x, y, a, h); this value is high when a track has just been initialized, has gone many frames without an update (due to occlusion), or has undergone irregular motion causing the Kalman filter to lose confidence in its estimate. The mean covariance is the trace normalized by the number of dimensions:

$$trace_i = \log(1 + \text{tr}(P_i)) \quad (15)$$

$$mean_i = \log \left(1 + \frac{\text{tr}(P_i)}{4} \right) \quad (16)$$

where P_i is the Covariance Matrix of the i^{th} track.

- Time since update: Unlike covariance trace (15) and covariance mean (16), "time since update" represents the number of consecutive frames during which a track relies solely on prediction without receiving an update from a detection. The higher this value, the greater the likelihood that the Kalman filter's prediction drifts from the actual position due to accumulated error across prediction steps. Therefore, "time since update" and covariance offer complementary perspectives for assessing the reliability of a track.

$$\Delta t_i = \max(0, t - f_i) \quad (17)$$

$$\tau_i = \min \left(\frac{\Delta t_i}{T_{max}}, 1 \right) \quad (18)$$

where t is the current frame id, f_i is the latest frame which is updated, Δt_i is the amount of frames were lost (were not updated). T_{max} is 300 that is simulated track will be deleted after 300 frames. τ_i is time since update normalization.

- Velocity Magnitude: It is the magnitude of the velocity vector estimated by the Kalman filter. It indicates how fast or slow the track is moving.

$$v_i = \tanh \left(\sqrt{vx_i^2 - vy_i^2} \right) \quad (19)$$

where vx_i^2 and vy_i^2 is the velocity of the bounding box's x and y coordinates as predicted by the Kalman Filter.

b) *Appearance features.*: Appearance features include cosine distance between track and detection embeddings, as well as embedding variance computed from the historical ReID features of the track.

- Cosine Distance: Measure the visual difference between track i and detection j based on embeddings extracted by the ReID model. The track maintains an embedding smoothed over time using EMA, whereas the detection has an embedding freshly extracted from the bounding box in the current frame.

$$\text{cosine}_{sim}(\mathbf{i}, \mathbf{j}) = \frac{\mathbf{i} \cdot \mathbf{j}}{\|\mathbf{i}\| \|\mathbf{j}\|} \quad (20)$$

$$\mathbf{d}_{cosine}(\mathbf{i}, \mathbf{j}) = \text{clip}(1 - \text{cosine}_{sim}(\mathbf{i}, \mathbf{j}); 0, 1) \quad (21)$$

- Feature Variance: while the cosine distance (21) answers the question "Are the track and the detection similar?", variance addresses the question "Is this track's appearance cue itself reliable?". If a track's embedding fluctuates significantly across recent frames due to factors such as occlusion, lighting changes, or shifts in viewing angle then the cosine distance calculated at the current frame is less reliable, regardless of whether the value is high or low, compared to a track with a stable embedding over an extended period.

$$\sigma_i^2 = \text{Var} \left(\{\mathbf{f}_i^{(1)}, \dots, \mathbf{f}_i^{(k)}\} \right) \in \mathbb{R}^D \quad (22)$$

where $\text{Var}(\cdot)$ computes the variance independently for each embedding dimension.

$$\text{variance}_i = \frac{1}{D} \sum_{d=1}^D \sigma_i^2[d]. \quad (23)$$

c) *Detection features.*: Detection features include detection confidence, bounding box area, and aspect ratio. Unlike motion features and appearance features, the three detection features depend solely on the detection d_j itself.

Let $w_j = x_2 - x_1$ and $h_j = y_2 - y_1$ denote the width and the height of the detection bounding box. The three features are defined as:

$$0 \leq \text{conf}_j \leq 1 \quad (24)$$

$$\text{bbox}_{\text{area}} = \frac{\log(1 + (w_j * h_j))}{15} \quad (25)$$

$$\text{bbox}_{\text{aspect}} = \frac{w_j}{h_j} \quad (26)$$

d) *Track history features.*: Track history features include track age, tracklet length, and a binary indicator showing whether valid appearance embeddings are available.

- Track age: It is a measure of the number of frames a track has existed, from its creation up to the present moment. A track that has persisted longer is considered more reliable than one that has just been generated.

$$\text{track}_{\text{age}} = \text{track}_{\text{curr}} - \text{track}_{\text{start}} \quad (27)$$

where $\text{track}_{\text{curr}}$ is current frame, $\text{track}_{\text{start}}$ is the frame of that track at the moment of its initial creation.

- Tracklet length: Unlike track age, tracklet length is reset to zero whenever a track is reactivated. It reflects short-term stability—distinguishing between a "long-standing track that has just suffered an interruption" and one that is "long-standing and continuously stable."

The variable ℓ_i denotes the most recent streak of successful matches.

$$\hat{\ell}_i = \min\left(\frac{\ell_i}{L_{\max}}, 1\right), \quad L_{\max} = 30 \text{ frames.} \quad (28)$$

- Appearance-availability indicator: This is a binary variable indicating whether the pair (track i , detection j) has valid embeddings. When ReID is unavailable (i.e., the track has never been matched based on appearance, or an embedding could not be extracted for the detection), the cosine distance (21) is assigned a neutral value of 0.5 and a variance (23) of 0.

$$\text{has}_{\text{appearance}} = \begin{cases} 1, & \text{if } \text{smooth_feat}_i \neq \emptyset \wedge \text{curr_feat}_j \neq \emptyset \\ 0, & \text{otherwise.} \end{cases} \quad (29)$$

Together, these features allow DMA to estimate not only the association cost, but also the reliability of each cost component.

IV. IMPLEMENTATION DETAILS

Our implementation is based on the official ByteTrack framework with YOLOX as the object detector. The original detection and BYTE association pipeline remain unchanged to ensure a fair comparison with the baseline, while the proposed LTC motion predictor and Dynamic Motion–Appearance (DMA) fusion are integrated as optional modules.

The LTC/CfC residual predictor is implemented in `yolox/tracker/ltc_motion.py`. Instead of replacing the Kalman Filter, the predictor estimates a residual motion correction from historical trajectories and refines the predicted state while preserving the efficiency and stability of the original motion model. The DMA module is implemented in `yolox/DMA/`, where `model.py` defines the DynamicWeightNet, `features.py` extracts a 15-dimensional feature vector for each track–detection pair, and `fuse.py` performs adaptive motion–appearance fusion.

DynamicWeightNet is implemented as a lightweight multi-layer perceptron consisting of two hidden layers (64 and 32 neurons), each followed by LayerNorm and ReLU activation. The input feature vector combines motion cues, appearance similarity, detection confidence, bounding-box attributes, and track-history information. The network predicts two adaptive fusion weights that are normalized by a softmax layer to balance motion and appearance costs during data association. During inference, when valid appearance embeddings are unavailable, the fusion module automatically falls back to motion-only association to avoid unreliable ReID costs.

V. EXPERIMENTS

A. Datasets and Metrics

We evaluate the proposed method on SportMOT and MOT17. SportMOT contains complex motion patterns and visually similar targets, making it suitable for evaluating adaptive motion–appearance association. MOT17 is a standard pedestrian tracking benchmark with crowded scenes and frequent occlusions.

We report HOTA, DetA, AssA, MOTA, and IDF1. HOTA balances detection and association accuracy, DetA measures detection accuracy, AssA measures association accuracy, MOTA summarizes tracking errors, and IDF1 evaluates identity preservation.

B. Experimental Protocol

All experiments are conducted under the same detector configuration as the original ByteTrack framework to ensure that performance differences mainly reflect the effectiveness of the proposed tracking modules. Unless otherwise specified, all hyperparameters remain identical to those of the ByteTrack baseline. Performance is evaluated using the standard TrackEval toolkit with the HOTA, DetA, AssA, MOTA, and IDF1 metrics.

The proposed LTC-DMA tracker is evaluated on two complementary benchmarks. SportMOT contains frequent player interactions, abrupt nonlinear motion, and severe appearance

TABLE I: Comparison with state-of-the-art methods on SportMOT. Higher values indicate better performance. Best results are shown in bold.

Method	HOTA	DetA	AssA	MOTA	IDF1
FairMOT	49.3	70.2	34.7	86.4	53.5
GTR	54.5	64.8	45.9	67.9	55.8
QDTrack	60.4	77.5	47.2	90.1	62.3
ByteTrack	62.1	76.5	50.5	93.4	69.1
CenterTrack	62.7	82.1	48.0	90.8	60.0
OC-SORT	68.1	84.8	54.8	93.4	68.0
BoT-SORT	68.7	84.4	55.9	94.5	70.0
TransTrack	68.9	82.7	57.5	92.6	71.5
DiffMOT	72.1	86.0	60.5	94.5	72.8
LTC-DMA (Ours)	78.35	88.32	69.52	98.12	82.60

TABLE II: Comparison with state-of-the-art methods on MOT17. Higher values indicate better performance. Best results are shown in bold.

Method	HOTA	DetA	AssA	MOTA	IDF1
DiffusionTrack	60.8	63.2	58.8	77.9	73.8
MotionTrack	61.6	—	60.2	78.6	75.1
ByteTrack	63.1	64.5	62.0	80.3	77.3
OC-SORT	63.2	63.2	63.4	78.0	77.5
MixSort-OC	63.4	63.8	63.2	78.9	77.8
MixSort-Byte	64.0	64.1	64.2	79.3	78.7
C-BIoU	64.1	64.8	63.7	79.7	79.7
StrongSORT	64.4	64.6	64.4	79.6	79.5
DiffMOT	64.5	64.7	64.6	79.8	79.3
Deep OC-SORT	64.9	64.1	65.9	79.4	80.6
SparseTrack	65.1	65.3	65.1	81.0	80.1
LTC-DMA (Ours)	78.19	79.15	77.71	90.32	87.63

ambiguity, making it particularly challenging for data association. MOT17 is a widely used pedestrian tracking benchmark featuring crowded scenes, partial occlusions, and diverse camera viewpoints.

C. Comparison with State-of-the-Art on SportMOT

As shown in Table I, the proposed LTC-DMA tracker consistently outperforms previous methods across all evaluation metrics. Compared with ByteTrack, our method achieves substantial improvements in HOTA and IDF1, demonstrating stronger association accuracy and identity preservation. These results indicate that nonlinear motion refinement and adaptive motion–appearance fusion complement each other effectively in challenging sports scenarios.

D. Comparison with State-of-the-Art on MOT17

Table II demonstrates that the proposed LTC-DMA framework also generalizes well to the MOT17 benchmark. Compared with ByteTrack, the proposed method achieves consistent improvements across HOTA, AssA, MOTA, and IDF1, indicating that adaptive motion modeling and context-aware fusion remain effective in crowded pedestrian tracking scenarios.

TABLE III: Ablation study on SportMOT.

Variant	HOTA	DetA	AssA	MOTA	IDF1
ByteTrack	68.8	80.6	58.8	97.5	77.7
ByteTrack + ReID (fixed $w_{reid} = 0.5$)	70.4	80.6	61.6	97.0	81.2
ByteTrack + ReID (BO tuned w_{reid})	78.4	88.4	69.6	98.2	83.3
ByteTrack + DMA (Ours)	73.0	81.2	65.6	97.8	83.7

TABLE IV: Ablation study on MOT17.

Variant	HOTA	DetA	AssA	MOTA	IDF1
ByteTrack	77.7	79.4	76.5	90.2	86.4
ByteTrack + ReID (fixed $w_{reid} = 0.5$)	75.3	78.6	72.6	89.5	82.5
ByteTrack + ReID (BO tuned w_{reid})	72.9	77.5	69.2	87.8	79.0
ByteTrack + DMA (Ours)	78.2	79.2	77.7	90.3	87.6

E. Ablation Study

To evaluate the contribution of each proposed component, we compare the original ByteTrack baseline with several variants that progressively incorporate appearance information and adaptive motion–appearance fusion. These experiments are designed to analyze the effectiveness of the proposed association strategy and its contribution to overall tracking performance.

Furthermore, since DMA replaces the fixed motion–appearance weighting mechanism with one that adaptively learns weights for each track–detection pair, we pose the question: can a static weight—if meticulously fine-tuned—achieve performance comparable to DMA? To establish the strongest possible static baseline, we search for a fixed fusion weight $w_{reid} \in [0, 1]$ that maximizes HOTA on the validation set using Bayesian Optimization [?], evaluating it over 20 trials, where each trial corresponds to a full tracker run using that specific fixed weight. This variant, denoted as ByteTrack + ReID (BO), represents the upper bound of performance achievable by a static, global weight; it is included in Table III alongside the default fixed-weight variant (0.5/0.5) and the DMA variant (dynamic, pair-wise weighting).

F. Qualitative Analysis

Figure 4 presents representative tracking results in challenging scenarios involving heavy occlusion, abrupt motion, and crowded interactions. Compared with the ByteTrack baseline, the proposed LTC-DMA tracker produces more stable trajectories and substantially fewer identity switches. The LTC motion predictor improves trajectory continuity under nonlinear motion, while DMA adaptively balances motion and appearance cues to maintain reliable associations after occlusion.

G. Runtime Analysis

The proposed framework preserves the online tracking pipeline of ByteTrack and introduces only lightweight additional computation. The LTC module performs residual motion correction only when sufficient trajectory history is available, while DMA employs a compact multilayer perceptron for adaptive motion–appearance fusion. Consequently, the primary computational overhead arises from ReID feature extraction

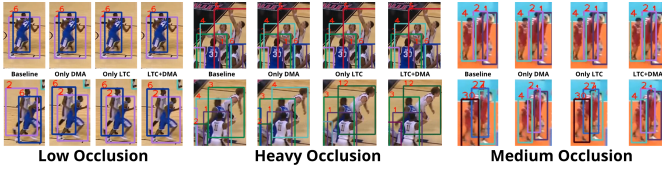


Fig. 4: Comparison of different association strategies under increasing occlusion severity. From left to right: low, heavy, and medium occlusion. Adaptive DMA fusion consistently improves tracking accuracy and identity preservation over fixed-weight association, with the performance gap becoming more evident as occlusion severity increases.

and pair-wise feature generation. A comprehensive runtime evaluation, including FPS, GPU memory consumption, and latency analysis, will be presented in future work.

VI. DISCUSSION

The experimental results demonstrate that adaptive motion–appearance fusion is more effective than using fixed global weights for data association. The reliability of motion and appearance cues varies significantly across targets and frames due to occlusion, abrupt motion, and appearance ambiguity. The proposed DMA module explicitly models these variations by exploiting pair-wise features that characterize motion uncertainty, appearance similarity, detection confidence, and track history. Consequently, the association process becomes more robust across diverse tracking scenarios.

The LTC residual predictor further complements DMA by improving motion estimation under nonlinear dynamics. Rather than replacing the Kalman Filter, LTC predicts a residual correction based on historical trajectories, allowing the tracker to better handle abrupt direction changes and irregular object motion while preserving the computational efficiency and stability of the original ByteTrack framework. The combination of refined motion prediction and adaptive association explains the consistent improvements observed on both SportMOT and MOT17.

Despite these promising results, several limitations remain. The proposed framework still relies on external ReID features, meaning that tracking performance may degrade when appearance embeddings are unreliable or unavailable. In addition, although the additional computational cost introduced by LTC and DMA is relatively small, ReID feature extraction remains the primary runtime bottleneck. Future work will investigate more efficient appearance representations and lightweight adaptive fusion strategies to further improve the trade-off between tracking accuracy and computational efficiency.

VII. LIMITATIONS

Although the proposed method shows strong results, several limitations remain. First, the additional ReID and DMA components introduce computational overhead compared with pure ByteTrack. Second, the current DMA model is trained separately from the full tracker, which may limit end-to-end optimization. Third, further evaluation is needed under

moving-camera and multi-camera settings. Finally, broader ablation studies are required to isolate the contributions of LTC prediction, ReID features, and DMA fusion.

VIII. CONCLUSION

This paper presented LTC-DMA, an adaptive extension of ByteTrack that improves multi-object tracking through two complementary components: an LTC/CfC-based residual motion predictor and a Dynamic Motion–Appearance (DMA) fusion module. Instead of replacing the original tracking pipeline, the proposed framework enhances motion estimation by learning residual trajectory corrections and adaptively balances motion and appearance cues during data association.

Extensive experiments on the SportMOT and MOT17 benchmarks demonstrate that the proposed approach consistently outperforms the ByteTrack baseline and achieves competitive performance against recent state-of-the-art trackers. In particular, the improvements in HOTA, MOTA, and IDF1 indicate enhanced trajectory continuity, association accuracy, and identity preservation in challenging scenarios involving nonlinear motion, heavy occlusion, and crowded interactions.

Future work will focus on improving computational efficiency by reducing the cost of appearance feature extraction, exploring end-to-end adaptive association frameworks, and extending the proposed motion modeling strategy to more diverse tracking scenarios and real-time applications.

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